ECE 4670 Spring 2011 Lab 6
Radio Transceiver Chip Sets and Sensor Node Radios

1 Introduction

For applications requiring wireless data capability, there are many chip sets to choose from. Most all of these options use carrier frequencies in the unlicensed industrial, scientific, and medical (ISM) bands[1], as listed in Table 1. In this lab you will be exposed to some of the various technologies available. In the United States the more popular bands are 433 MHz, 915 MHz, and 2.450

Table 1: ISM bands around the world.

Frequency range [Hz]	Center frequency [Hz]	Availability
6.765–6.795 MHz	6.780 MHz	Subject to local acceptance
13.553–13.567 MHz	13.560 MHz	
26.957–27.283 MHz	27.120 MHz	
40.66–40.70 MHz	40.68 MHz	
433.05–434.79 MHz	433.92 MHz	Region 1 only and subject to local acceptance
902–928 MHz	915 MHz	Region 2 only
2.400–2.500 GHz	2.450 GHz	
5.725–5.875 GHz	5.800 GHz	
24–24.25 GHz	24.125 GHz	
61–61.5 GHz	61.25 GHz	Subject to local acceptance
122–123 GHz	122.5 GHz	Subject to local acceptance
244–246 GHz	245 GHz	Subject to local acceptance

GHz. Hands-on exposure to 433 MHz and 2.450 GHz devices will be the focus of this lab.

2 The Hope RF 12B 433 MHz FSK Transceiver

[1] http://en.wikipedia.org/wiki/ISM_band

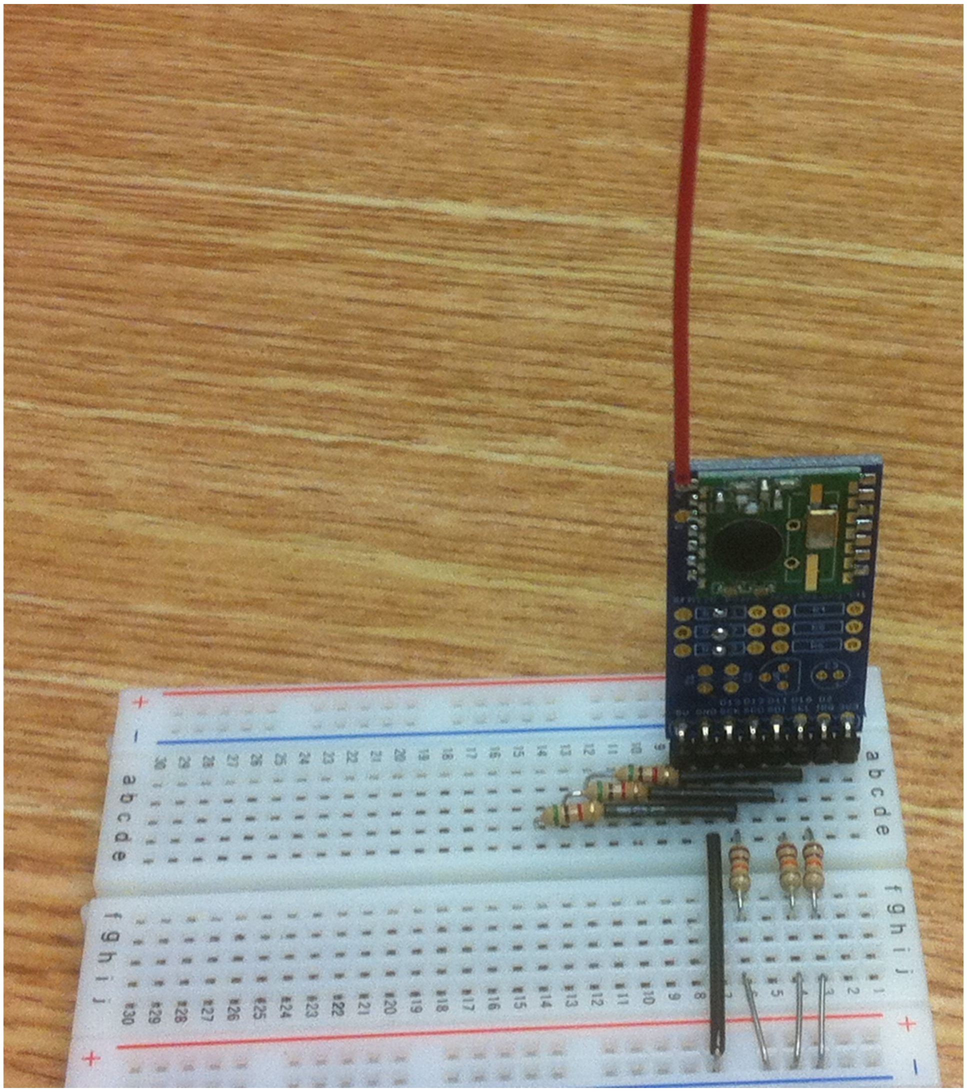

Figure 1: The Hope RF12B module mounted on a breakout board plugged into a breadboard.

3 The Digi XBee Series 1 802.15.4 Sensor Node Radio

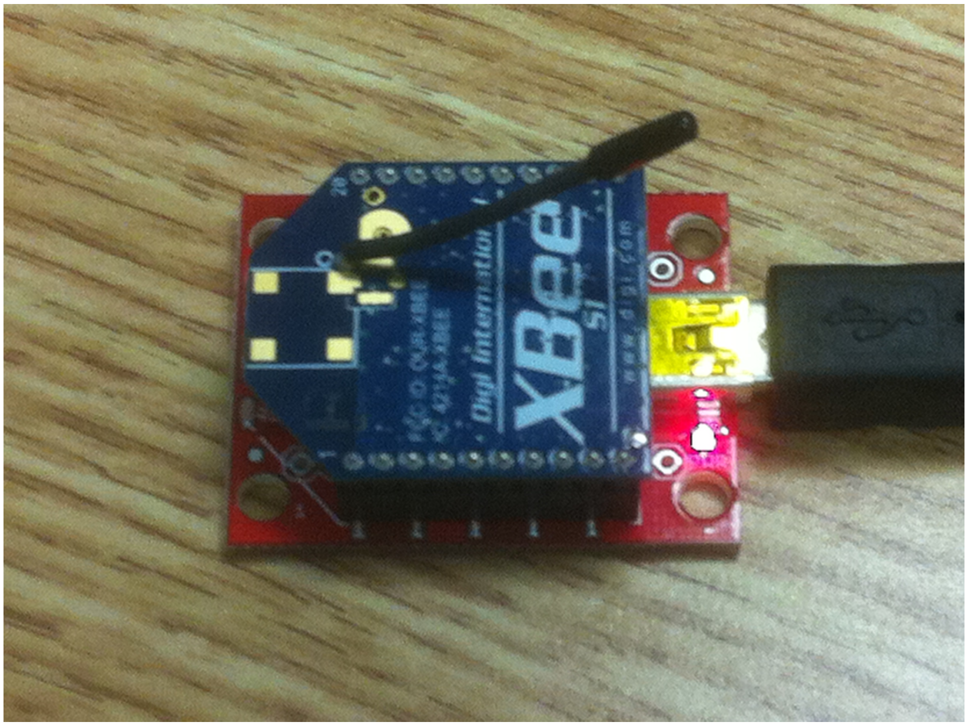

Figure 2: XBee series 1 802.15 2.4 GHz sensor node radio in the Explorer breakout board.

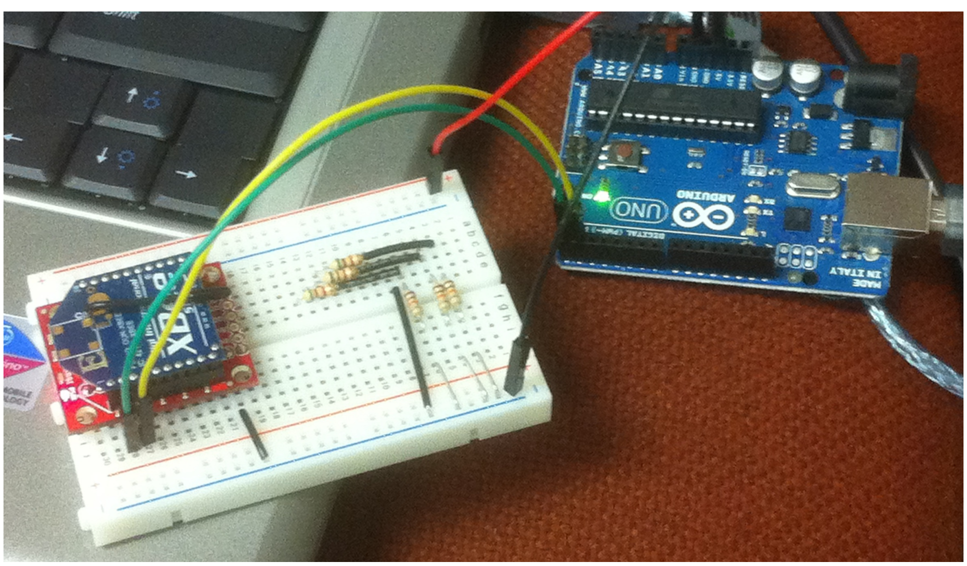

Figure 3: XBee series 1 802.15 2.4 GHz sensor node radio in a breakout board interfaced to an Arduino microcontroller.